

Daily Challenge: Understanding The Essence Of Machine Learning

Project Blueprint: Student success Predictor

1. Problem & ML Type for A Student success Predictor

- Goal: Predict if a student will Pass or Fail their final exam.
- ML Type Chosen: Supervised Learning
- Justification: We have a labeled target (exam outcome) to train the model, making this a classification problem.

2. Core Features

Feature	Why It's Powerful
1. Assignment Submission Rate	Low submission rates indicate disengagement or difficulty grasping material.
2. Time Spent on Difficult Topics	Struggling students may spend excessive time or avoid challenging topics entirely.
3. Forum Participation	Active discussion correlates with engagement; silence may signal confusion.

Quiz scores, login frequency, video lecture completion rate. are also available features.

3. Model Type

Model Family Chosen: Classification Models because the goal is to predict a binary outcome (Pass/Fail), not a continuous value.

4. Evaluation Strategy

The More Critical Error: Mistake A (False Negative)

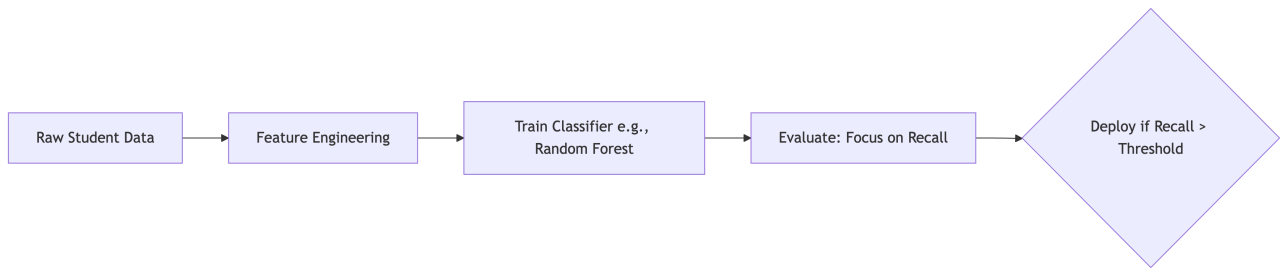
>> Predicting a student will pass when they actually fail (FN) denies them intervention, harming their academic outcome and the platform's reputation.

A False Positive (unnecessary help) is less costly than a missed at-risk student.

Considerations

- **Data Source:** platform logs (assignments, clicks, forum posts).

- **Model Tuning:** prioritize high recall to minimize False Negatives.
- **Ethics:** Avoid bias (e.g., penalizing students with slow internet for low activity).



Result : technical rigor + practical impact = ensuring proactive student support